Advanced Computer-Aided VLSI System Design

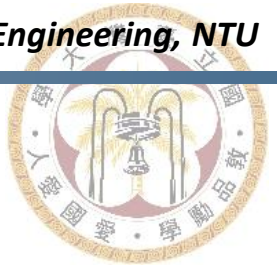
Chapter 7. Cell-Based APR Flow Part2

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Graduate Institute of Electronics Engineering, National Taiwan University



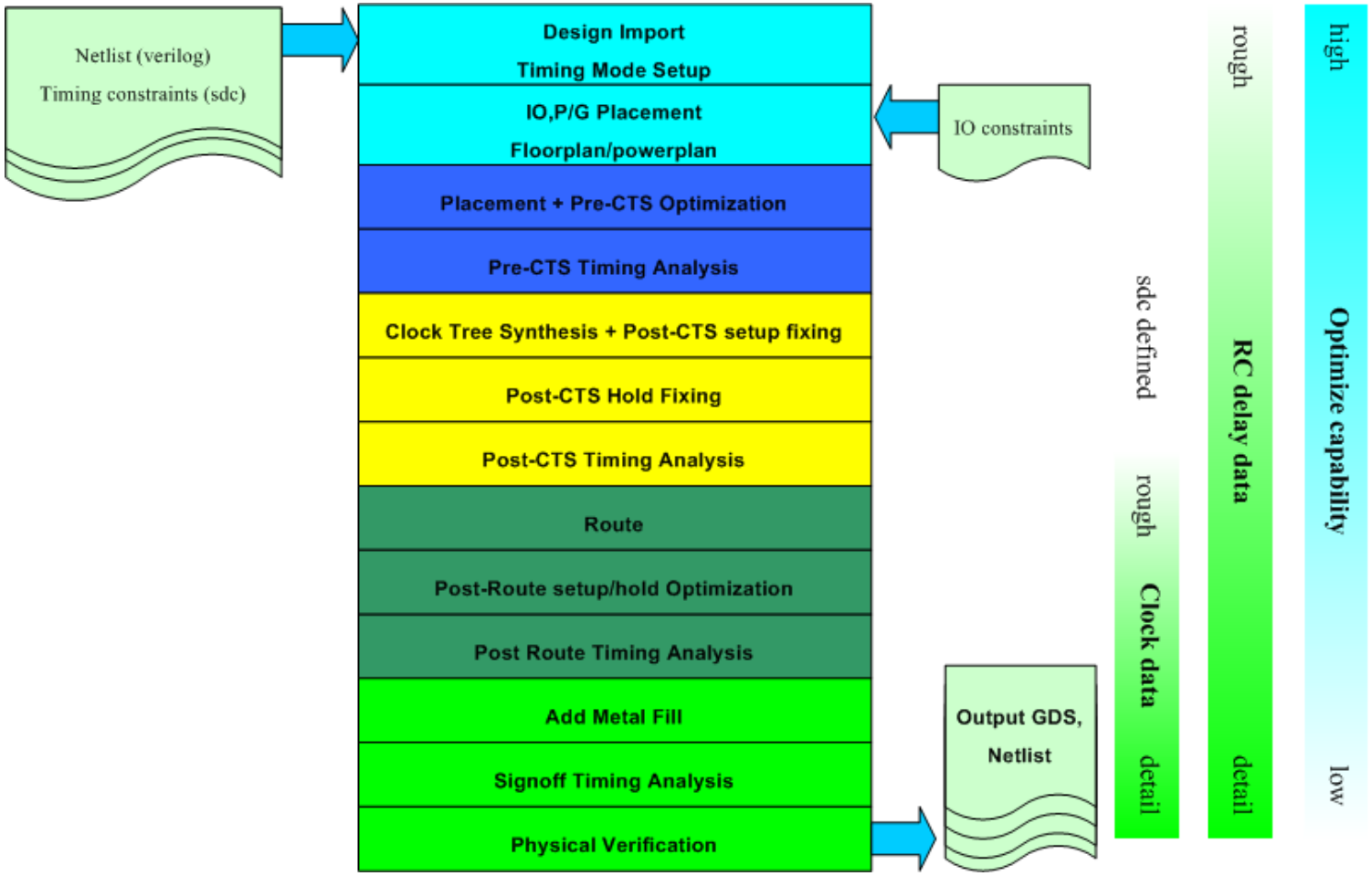
NTU GIEE



Outline

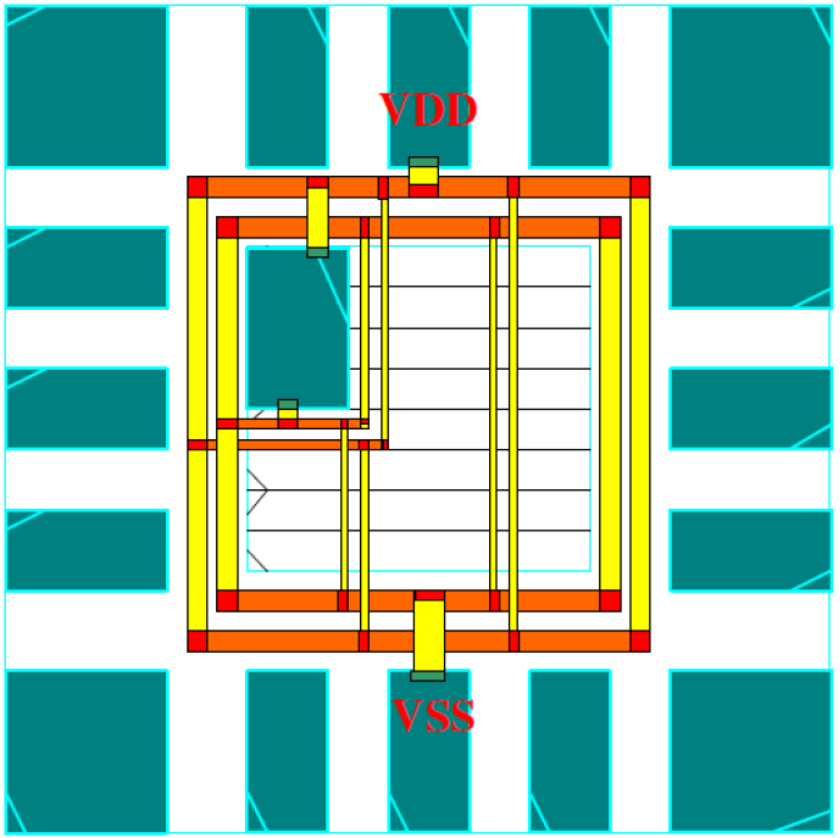
- Automatic Place & Route – Innovus
  - Power Planning
  - Placement
  - Clock Tree Synthesis
  - Routing
  - Output Data
- Common DRC Errors

Innovus APR Flow



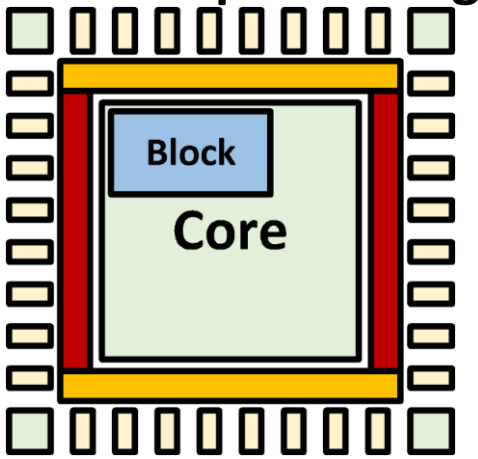
Power Planning Order

- **Concept: connect wider nets prior then narrow ones**
  - Create power ring
  - Connect pad pin
  - Create block ring
  - Connect block pin
  - Create stripe
  - Connect follow pin

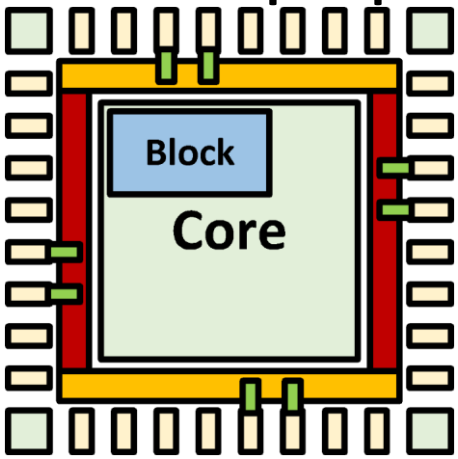


Power Planning Overview

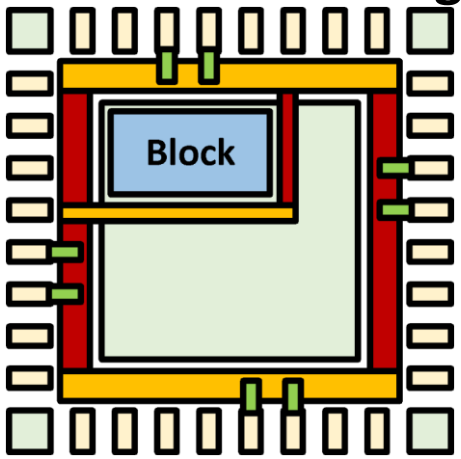
1. Create power ring



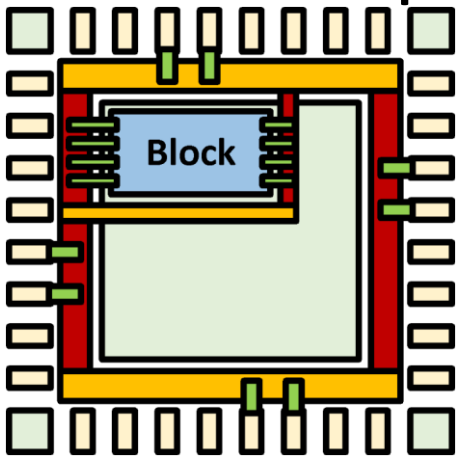
2. Connect pad pin



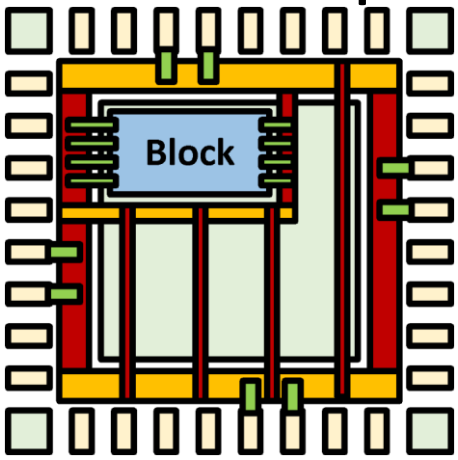
3. Create block ring



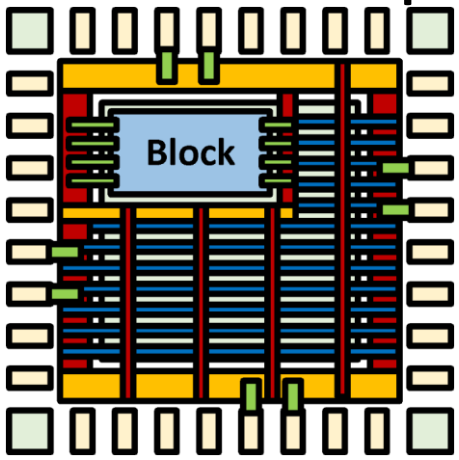
4. Connect block pin



5. Create stripe

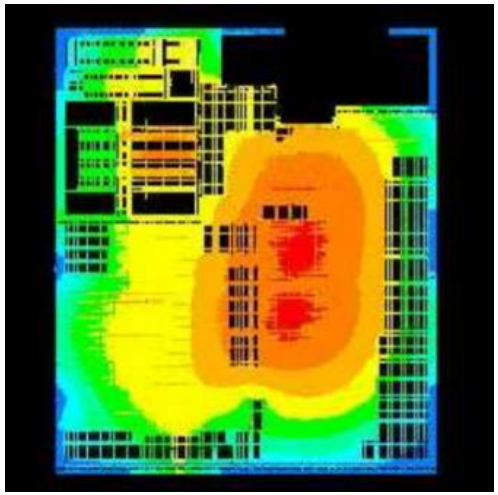


6. Connect follow pin

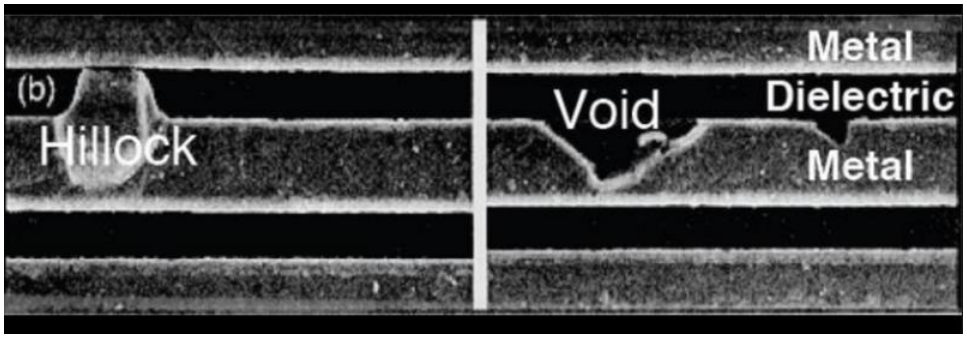


Power Planning Challenges in Advanced Nodes

- As technology nodes scale down,
  - Wires become thinner; cells become smaller but denser
  - **IR drop** increases
  - **Electromigration** worsens
- Proper power planning is essential to deal with these issues.



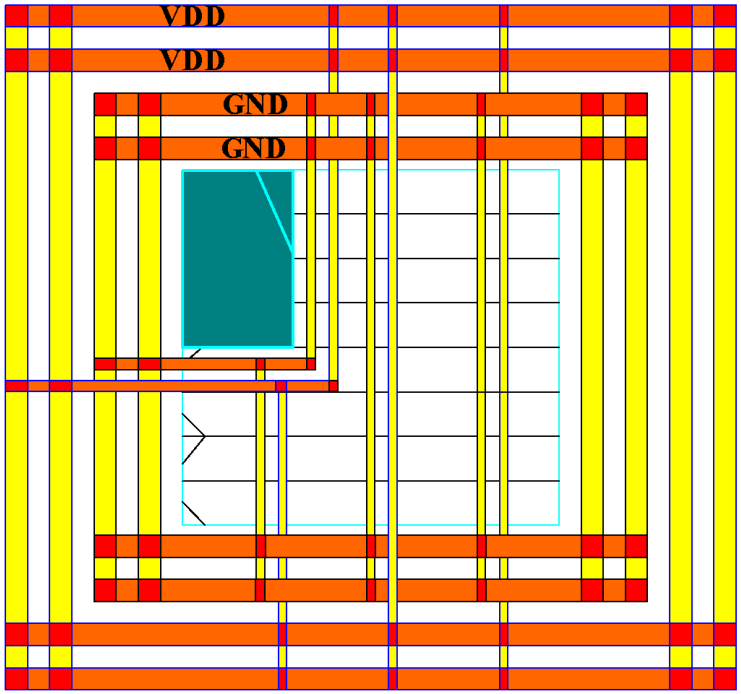
IR Drop

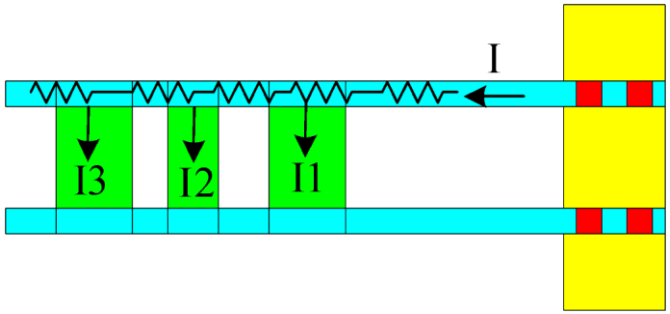


Electromigration

Power Planning: Stripe

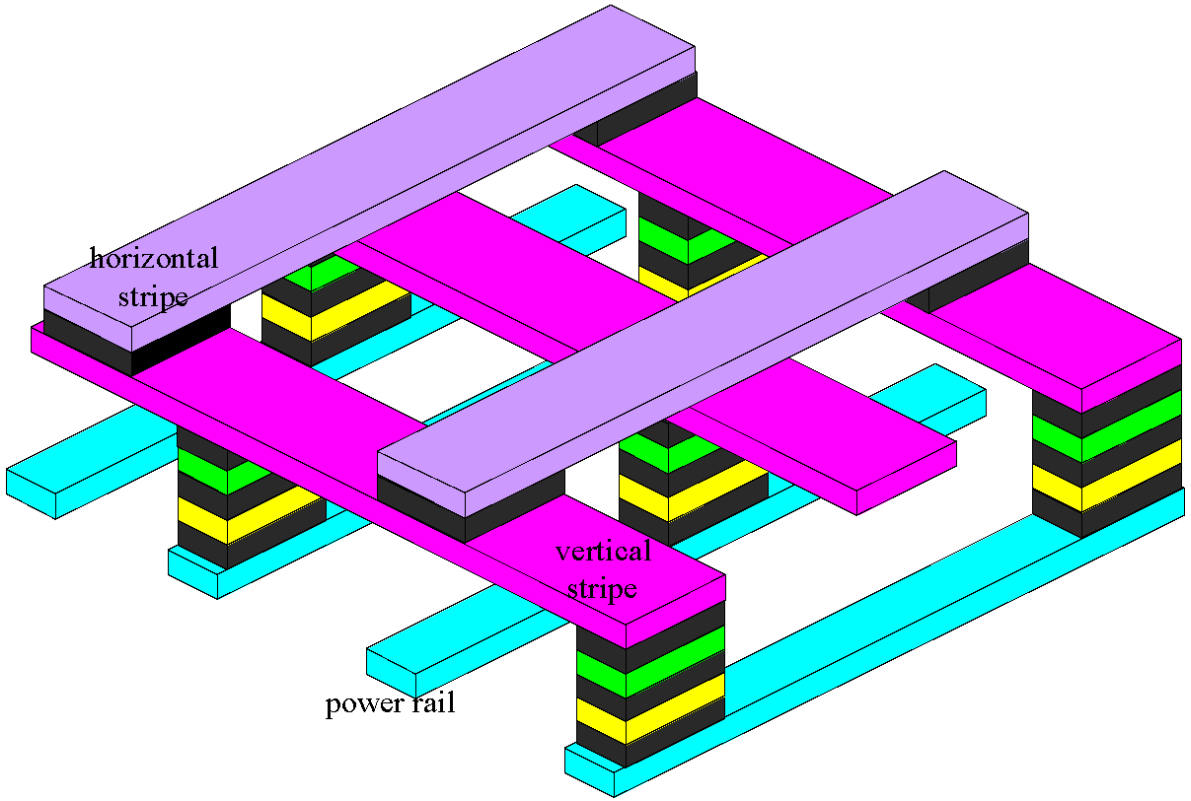






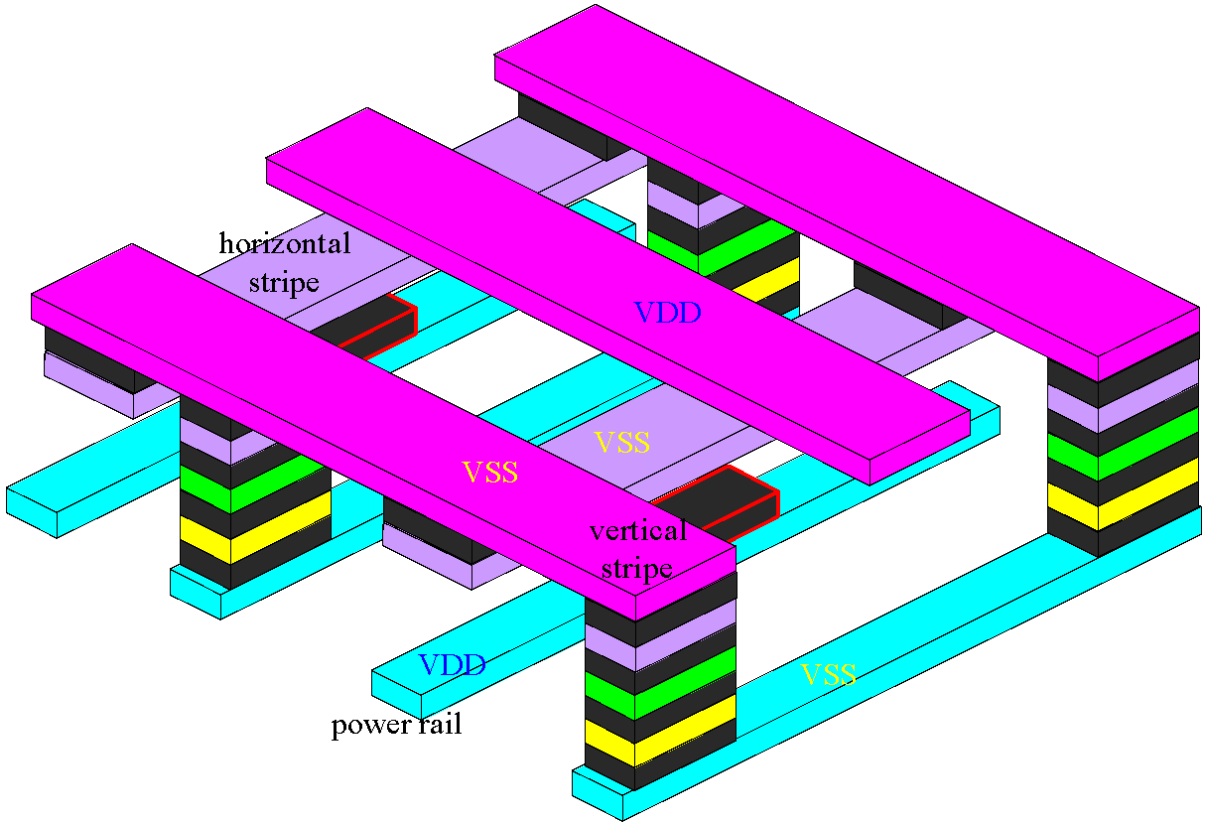
Power Planning: Stripe

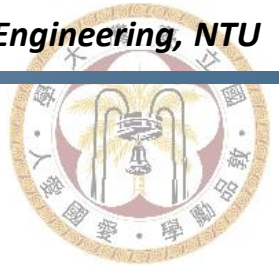
- Avoid to use wire group when creating stripes
- Connect follow pin only with lowest stripe
- The lowest stripe must be vertical



Power Planning: Stripe

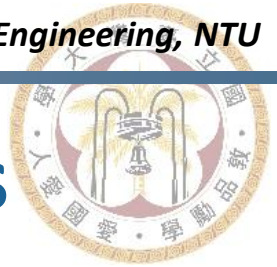
- If the lowest stripe is created with horizontal layer, the follow pin may fail to connect to the stripe





Power Stripe Strategy

- The density of power mesh dominate the capability of current transfer
- Width & Set to Set distance
  - Used to adjust power plan density
  - Thinner, more frequent stripes are better than thicker, less frequent stripe
- Stripes cut back on routing resource
  - Make stripe every metal above metal 4
- Let follow pin only connect to the lowest stripe



Early Rail Analysis: Power Analysis

- Prerequisite: Signal toggle count format (tcf) or waveform (vcd) from gate-level simulation
  - Or directly use default toggle rate

```
set_power_analysis_mode -method static
set_power_analysis_mode -corner max
set_power_analysis_mode -create_binary_db true
set_power_analysis_mode -write_static_currents true
set_power_analysis_mode -honor_negative_energy true
set_power_analysis_mode -ignore_control_signals true

set_default_switching_activity -reset
set_default_switching_activity -input_activity 0.2 -period 8.0
read_activity_file -reset
read_activity_file -format TCF -scope CHIP ./sim/pre_sim/CHIP.tcf
```

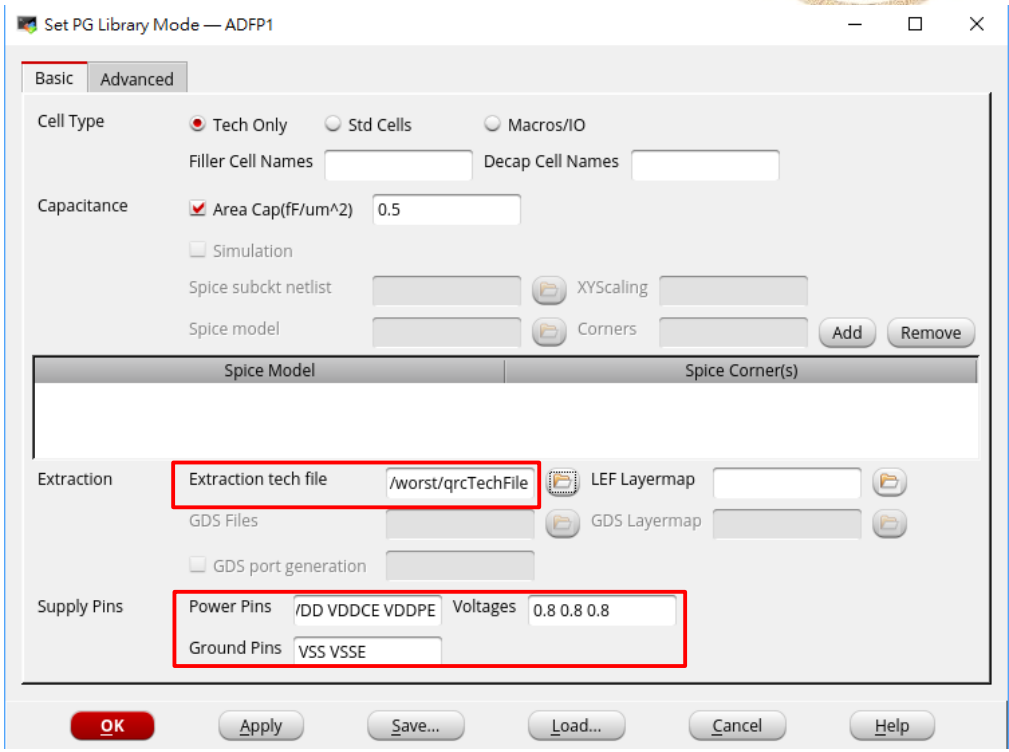
Default toggle rate

Toggle rate from gate-sim

- innovus > report_power
  - Output file: static_VDD.ptiavg and static_VSS.ptiavg

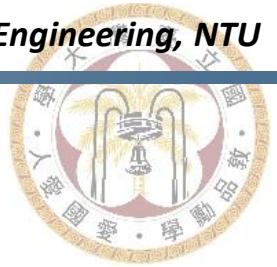
Early Rail Analysis: PG Library

- Set PG Library Mode
  - Extraction file: worst QRC



- Generate PG Library
  - Output file: techonly.cl





Early Rail Analysis

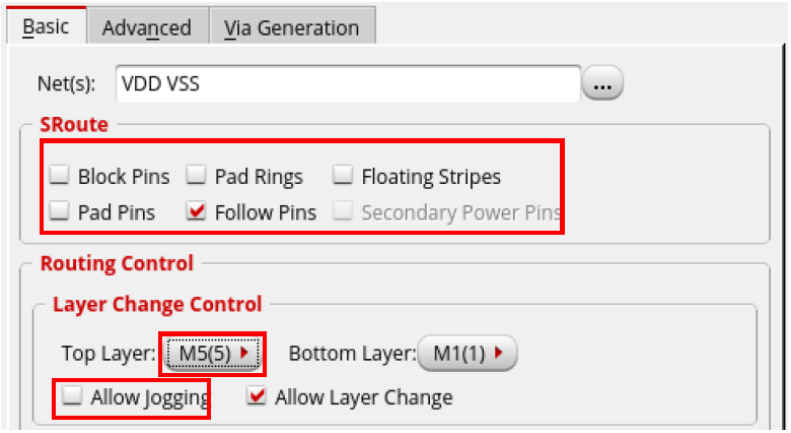
- Run rail analysis

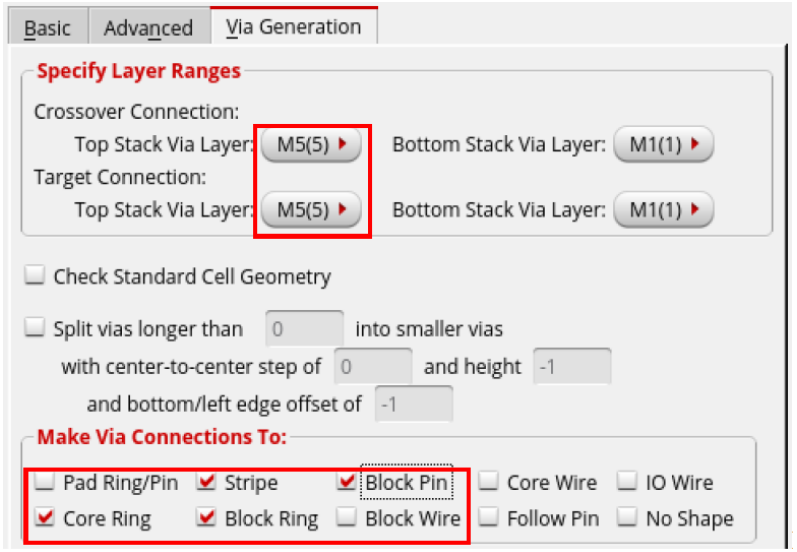
```
…
create_power_pads -net VDD -auto_fetch
create_power_pads -net VDD -vsrc_file CHIP_VDD.pp
create_power_pads -net VSS -auto_fetch
create_power_pads -net VSS -vsrc_file CHIP_VSS.pp
set_pg_nets -net VDD -voltage 0.8 -threshold 0.76
set_pg_nets -net VSS -voltage 0 -threshold 0.04
set_rail_analysis_domain -name PD -pwrnets { VDD} -gndnets { VSS}
set_power_data -reset
set_power_data -format current -scale 1 {static_VDD.ptiavg static_VSS.ptiavg}
set_power_pads -reset
set_power_pads -net VDD -format xy -file CHIP_VDD.pp
set_power_pads -net VSS -format xy -file CHIP_VSS.pp
set_package -reset
set_package -spice {} -mapping {}
set_net_group -reset
set_advanced_rail_options -reset
set_db power_grid_libraries techonly.cl
analyze_rail -type domain -results_directory ./ PD
```

- IR drop result: **PD_25C_avg_1/Reports/report_generation.log**
  - If IR drop result is acceptable (<3%), can connect follow pin

Power Planning: Add M2 Follow Pin

- Special Route (SRoute) to route M1 follow pin
- Method to reduce IR drop of follow pin: add M2 follow pin
  - Duplicate M1 follow pin
  - Add stripe over exist M1 follow pin





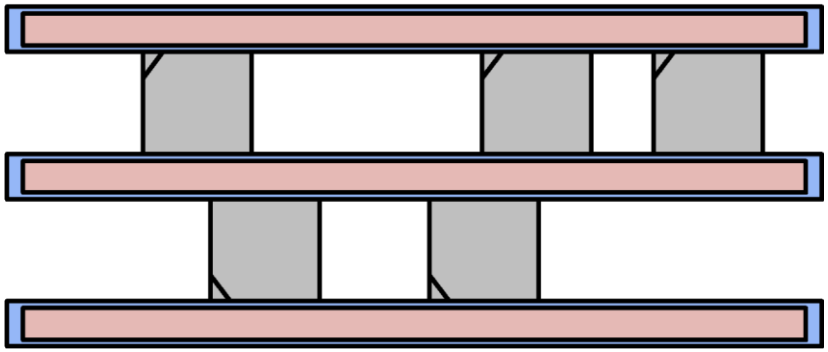
Power Planning: Add M2 Follow Pin

- Duplicate M1 follow pin

```
editSelect -net {VDD} -shape followpin
setEditMode -layer_maximum M1
editDuplicate -layer_horizontal M2 -layer_vertical M2
editResize -to 0.0x -direction y -side high
```

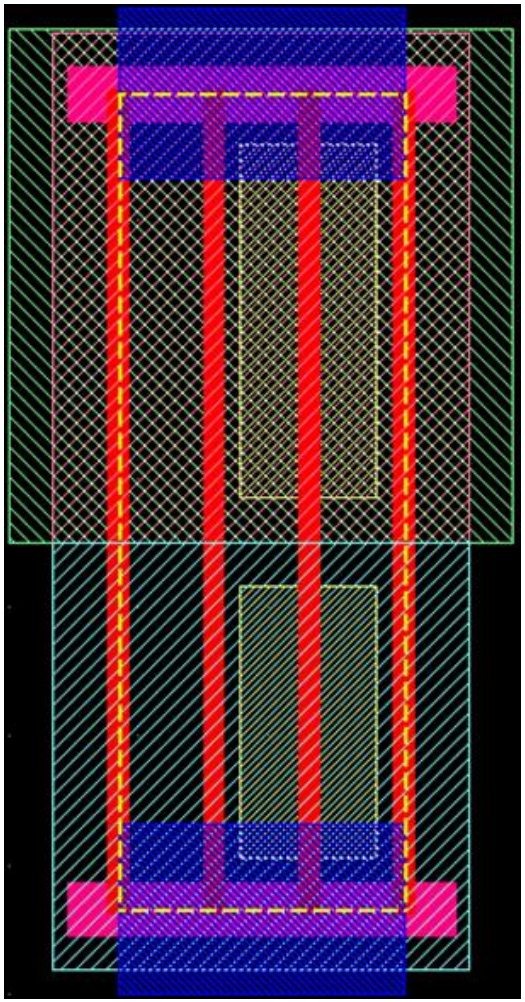
- Add stripe over exist M1 follow pin

```
addStripe -nets {VDD} -layer M2 -direction horizontal -width 0.0x \
-spacing 0 -set_to_set_distance [2 * row_height] \
-start from bottom -start_offset 0.0x …
```



Endcap Cell (Boundary Cell)

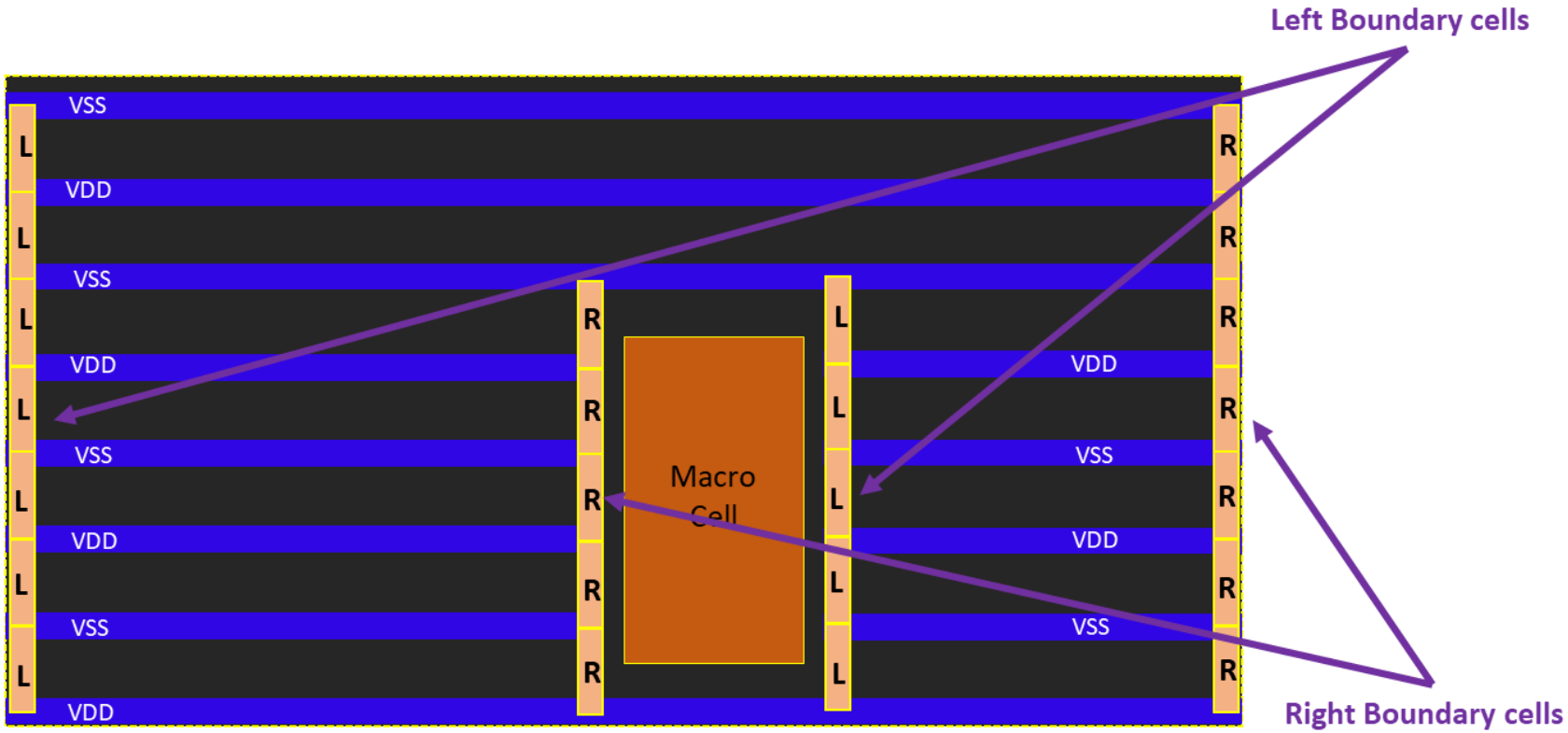
- To protect the gate of a standard cell placed near the boundary from damage during manufacturing
- The boundary cell is a physical-only cell, has no logical functions and therefore these cells are not a part of the netlist



Source: https://teamvlsi.com

Insert Endcap Cell (40nm, 28nm)

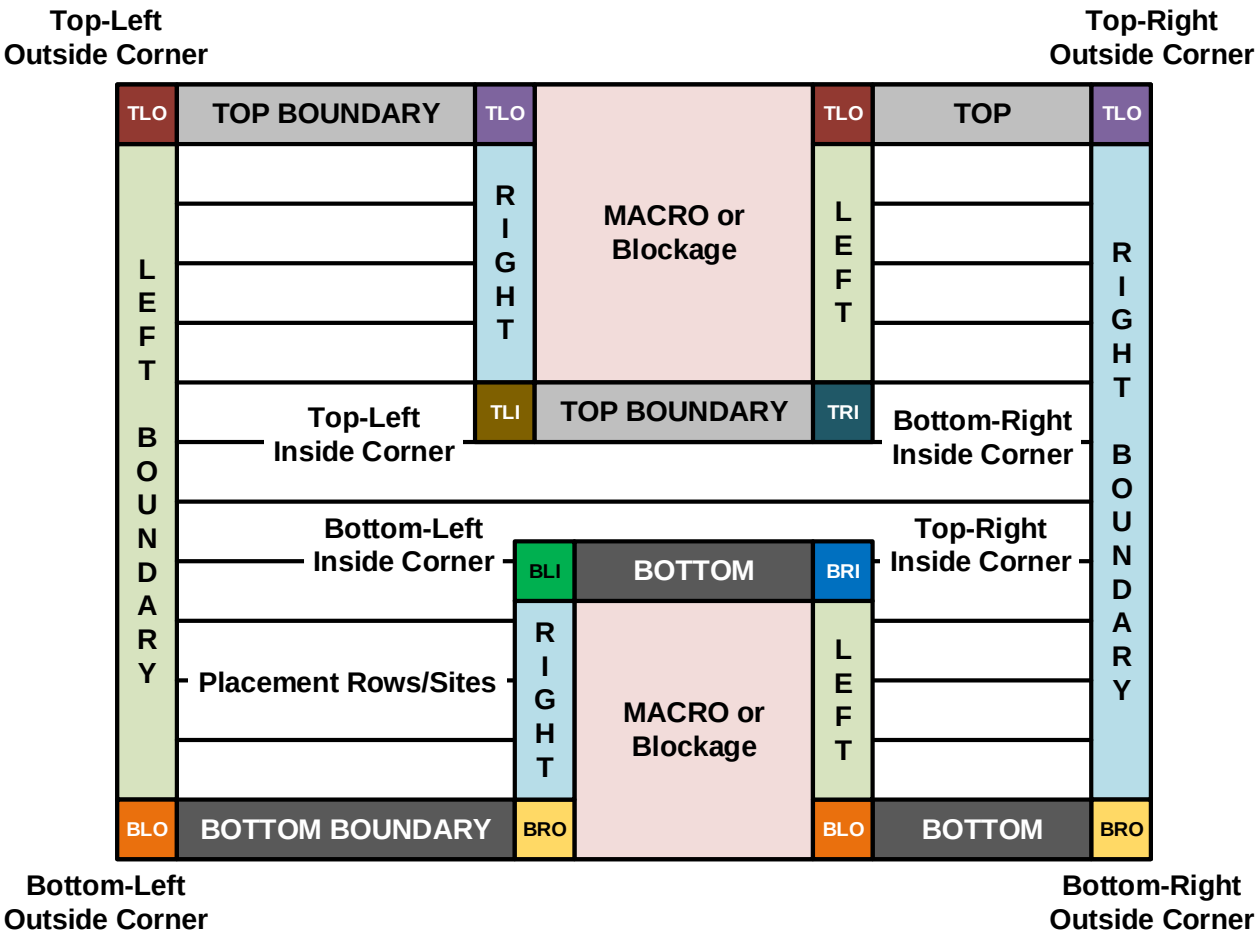
- Endcap cells are placed at the both ends of cell row for boundary DRC cleanup



Source: https://teamvlsi.com

Insert Endcap Cell (16nm, ADFP)

- Endcap cells are placed at all the boundaries for boundary DRC cleanup



Different Endcap for Each Boundary in ADFP

- Use correct endcap cells in different boundaries to prevent DRC

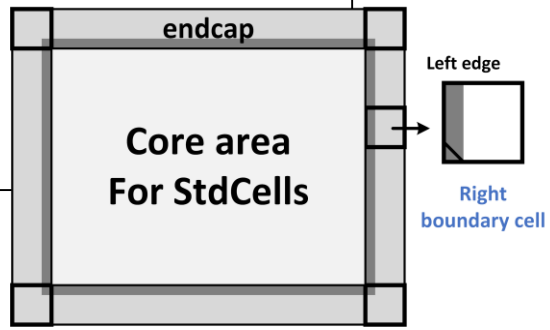
```
set_db add_endcaps_left_edge  {BOUNDARY_RIGHTBWP20P90LVT}
set_db add_endcaps_right_edge {BOUNDARY_LEFTBWP20P90LVT}

set_db add_endcaps_left_top_corner {BOUNDARY_PCORNERBWP20P90LVT}
set_db add_endcaps_left_bottom_corner {BOUNDARY_NCORNERBWP20P90LVT}

set_db add_endcaps_top_edge  {BOUNDARY_PROW1BWP20P90LVT \
BOUNDARY_PROW2BWP20P90LVT BOUNDARY_PROW3BWP20P90LVT \
BOUNDARY_PROW4BWP20P90LVT}
set_db add_endcaps_bottom_edge {BOUNDARY_NROW1BWP20P90LVT \
BOUNDARY_NROW2BWP20P90LVT BOUNDARY_NROW3BWP20P90LVT \
BOUNDARY_NROW4BWP20P90LVT}

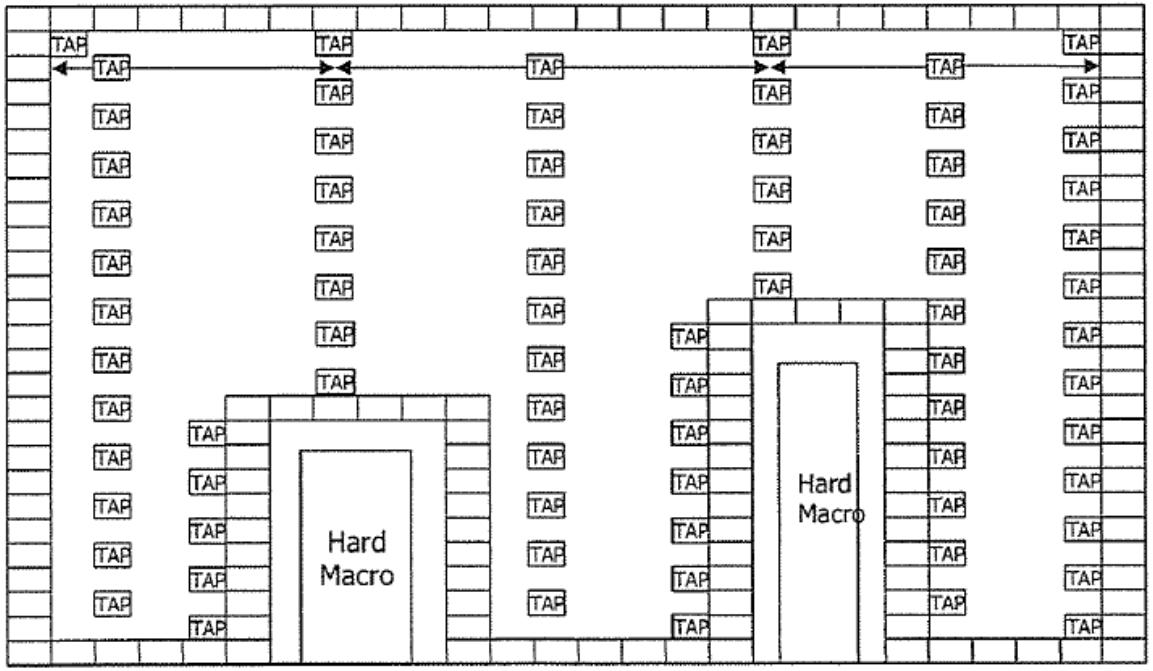
set_db add_endcaps_left_top_edge  {FILL3BWP20P90LVT}
set_db add_endcaps_right_top_edge {FILL3BWP20P90LVT}

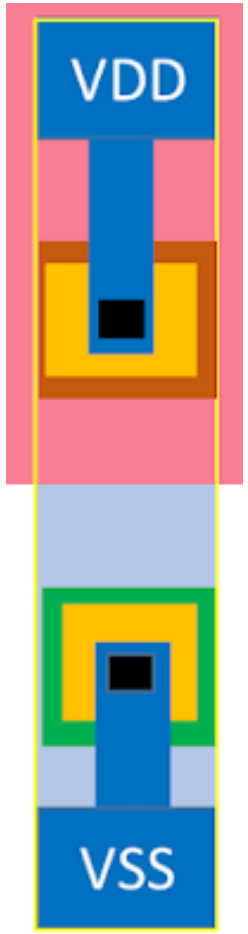
set_db add_endcaps_left_bottom_edge  {FILL3BWP20P90LVT}
set_db add_endcaps_right_bottom_edge {FILL3BWP20P90LVT}
```



Well Tap Cell

- It connects the NWELL to VDD and p-substrate to VSS in order to prevent the latch-up issue
- It has no input and output pins, therefore it is called a physical-only cell



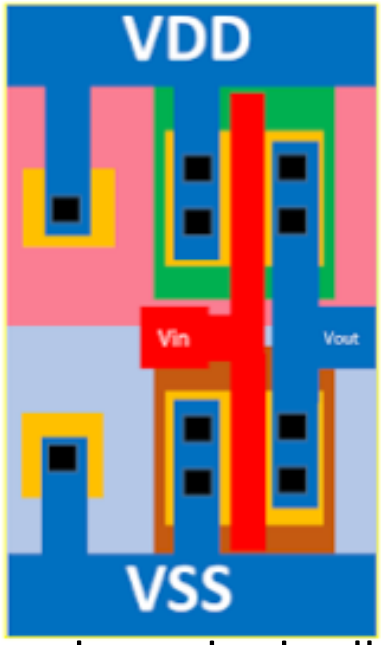


Source: https://teamvlsi.com

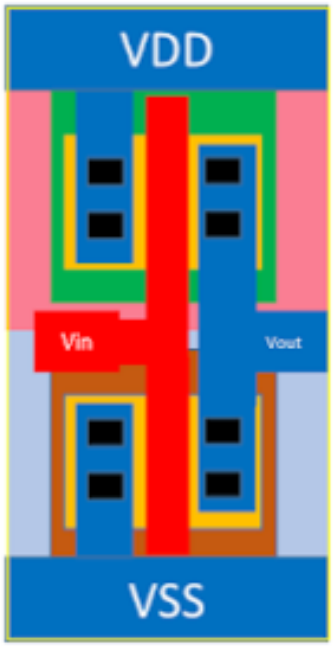
Insert Well Tap Cell

- Tapless structure has smaller area
- One well tap is shared by multiple standard cells within certain distance

```
set_db add_well_taps_insert_cells  {TAPCELLBWP20P90LVT rule 50.76}
addWellTap -checkerBoard
```



Traditional standard cell layout

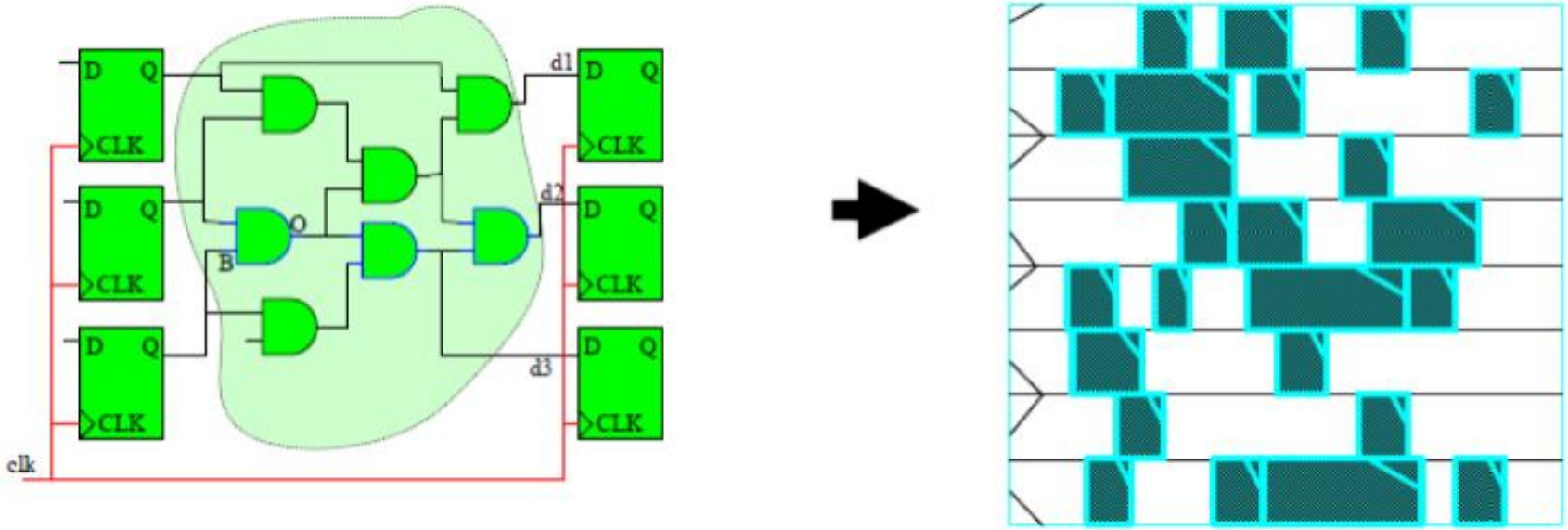


Tapless standard cell layout

Source: https://teamvlsi.com

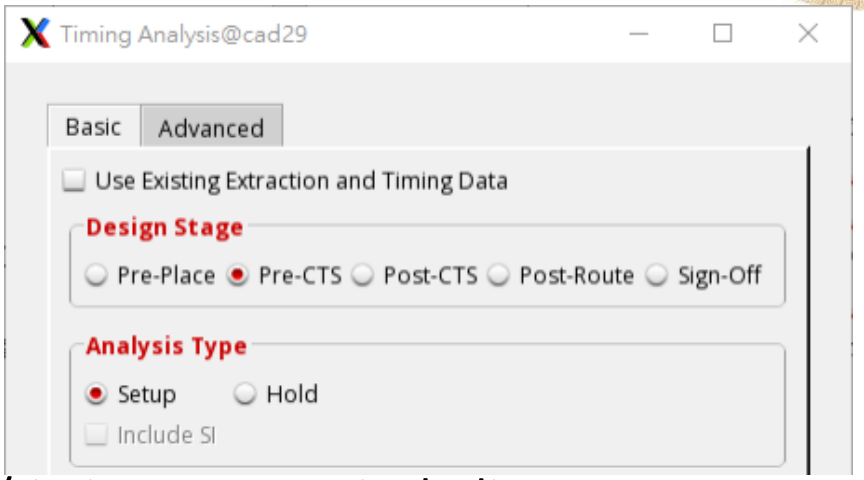
Placement

- *Place -> Place Standard Cells*
  - `Innovus #> createBasicPathGroups -expanded`
  - `Innovus #> get_path_groups`
  - `Innovus #> place_opt_design`
- **Using commands in placement**



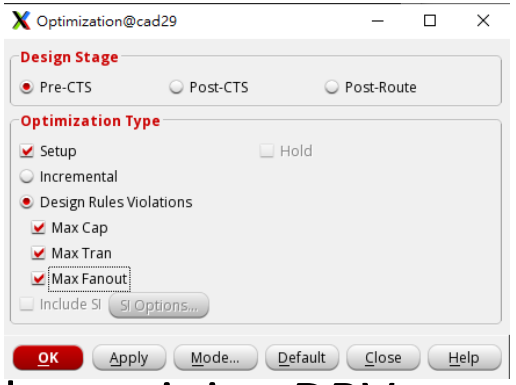
Pre-CTS Timing Analysis

- **To check timing**
  - ***Timing -> Report Timing***
- **Operations**
  - Trial route
  - RC extraction
  - Timing analysis
  - Generate reports are saved in ./timingReports , including
    - _preCTS.cap
    - _preCTS.fanout
    - _preCTS.tran
    - _preCTS_all.tarpt
- **Reports**
  - Timing summary
  - If the timing is met, pre-CTS optimization can be skipped



Pre-CTS Optimization

- **To optimize timing placed design for the first time with ideal clocks**
  - ***ECO -> Optimize Design***



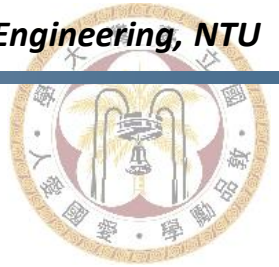
- **Operation**
  - Repair setup slack, setup times
  - Repair design rule violations (DRVs) and remaining DRVs

```
------------------------------------------------------------
     optDesign Final Summary
------------------------------------------------------------

Setup views included:
 av_func_mode_max

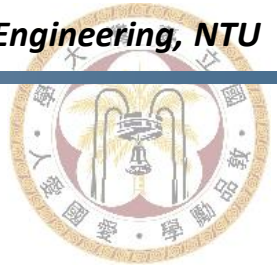
+--------------------+---------+---------+---------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | in2reg  | reg2out | in2out  | default |
+--------------------+---------+---------+---------+---------+---------+---------+
|           WNS (ns):|  0.005  |  0.005  |  3.491  |  0.235  |   N/A   |  0.000  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |  0.000  |   N/A   |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |    0    |   N/A   |    0    |
|         All Paths: |  2281   |  1877   |  771    |   13    |   N/A   |    0    |
+--------------------+---------+---------+---------+---------+---------+---------+

+----------------+-------------------------------+------------------+
|                |             Real              |       Total      |
|    DRVs        +---------------+---------------+------------------|
|                | Nr nets(terms)|  Worst Vio    | Nr nets(terms)   |
+----------------+---------------+---------------+------------------+
|   max_cap      |     0 (0)     |    0.000      |      0 (0)       |
|   max_tran     |     0 (0)     |    0.000      |      0 (0)       |
|   max_fanout   |     0 (0)     |      0        |      1 (1)       |
|   max_length   |     0 (0)     |      0        |      0 (0)       |
+----------------+---------------+---------------+------------------+
```



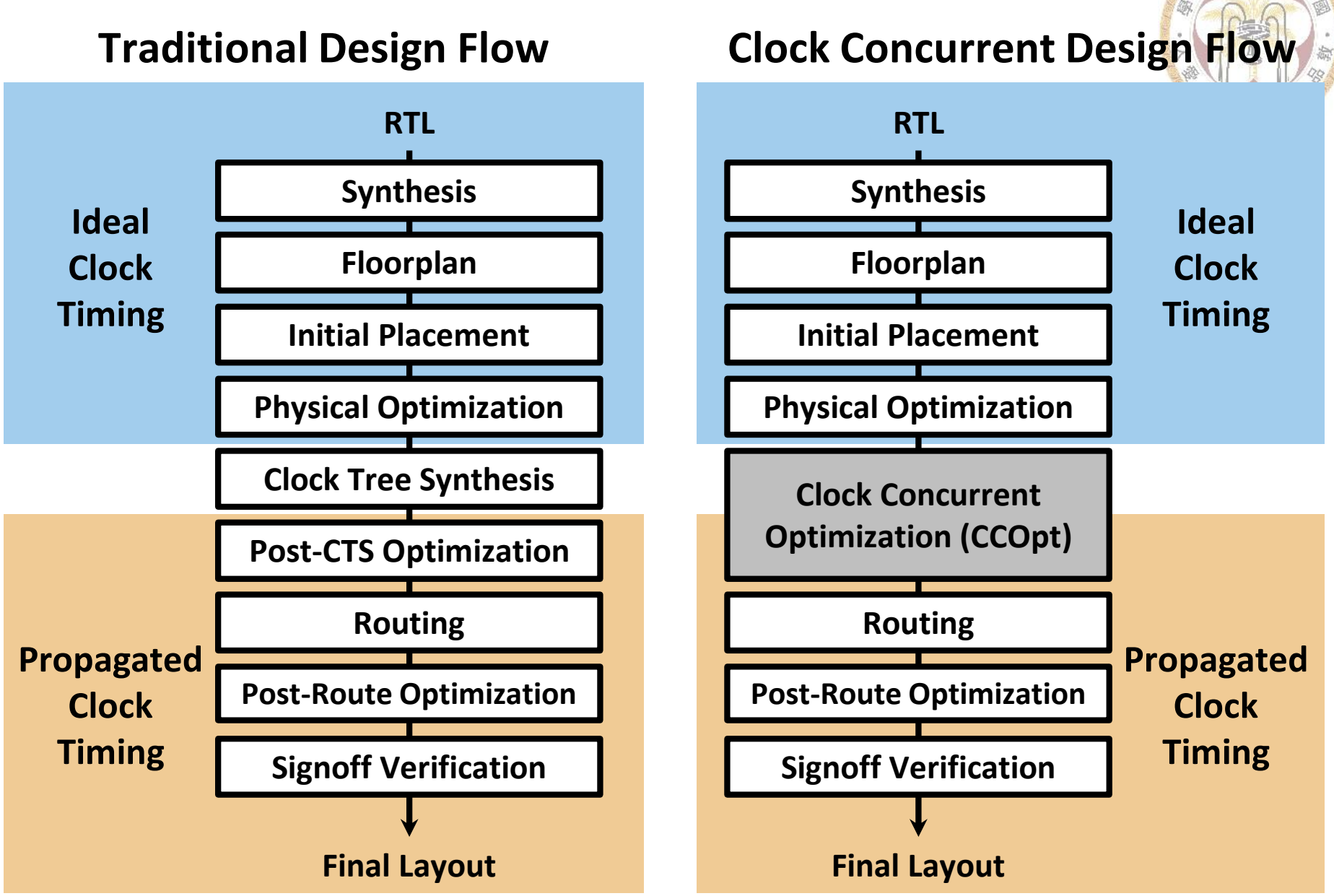
Clock Tree Synthesis (CTS)

- **Add multi-level buffer trees according to the clk specification**
- **A good clock tree should have the same feature as described in the clock estimation**
- **CTS result will not follow spec if clk estimation is not realistic**
- **Clock issues**
  - Heavy clock net loading
  - Long clock insertion delay
  - Clock skew
  - Clock to signal coupling effect
  - Clock is power hungry



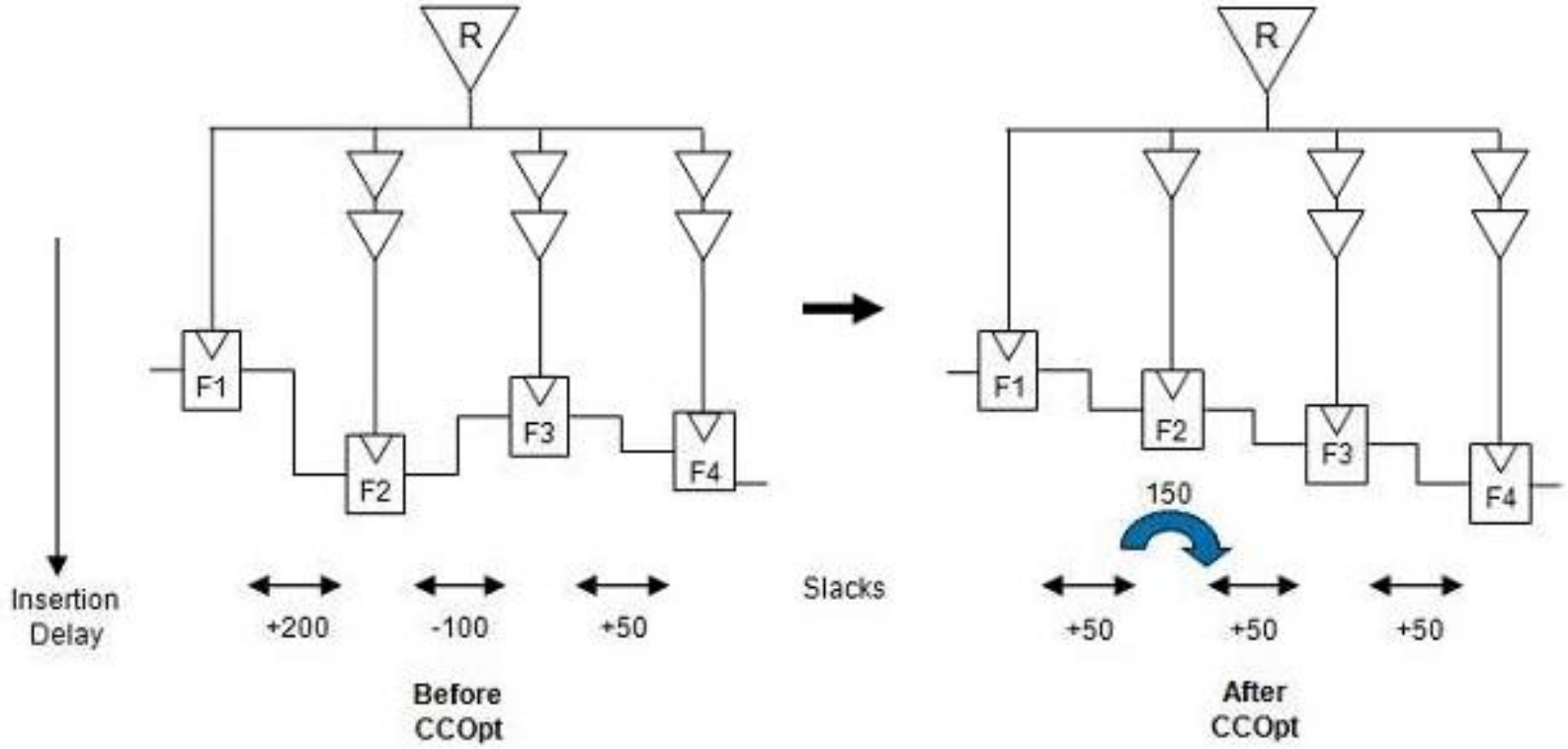
Edit .sdc before CTS

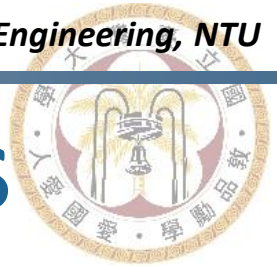
- **set_clock_latency**
  - source latency
  - clock network delay
- **set_clock_uncertainty**
  - clock jitter
  - clock skew
  - timing margin
- **Update .sdc file before CTS**
  - Comment **set_clock_latency**
  - Comment **set_clock_uncertainty**
  - Comment **set_ideal_network**
  - **Innovus #> update_constraint_mode -name CM_mode -sdc_files post_cts.sdc**


Traditional Design Flow
RTL
Synthesis
Floorplan
Initial Placement
Physical Optimization
Clock Tree Synthesis
Post-CTS Optimization
Routing
Post-Route Optimization
Signoff Verification
Final Layout
Ideal Clock Timing
Propagated Clock Timing
Clock Concurrent Design Flow
RTL
Synthesis
Floorplan
Initial Placement
Physical Optimization
Clock Concurrent Optimization (CCOpt)
Routing
Post-Route Optimization
Signoff Verification
Final Layout
Ideal Clock Timing
Propagated Clock Timing

Clock Concurrent Optimization (CCOpt)

- CCOpt is used to optimize the clock network by
  - Reasonable use of skewing
  - Simultaneous optimization of clock path and logic path



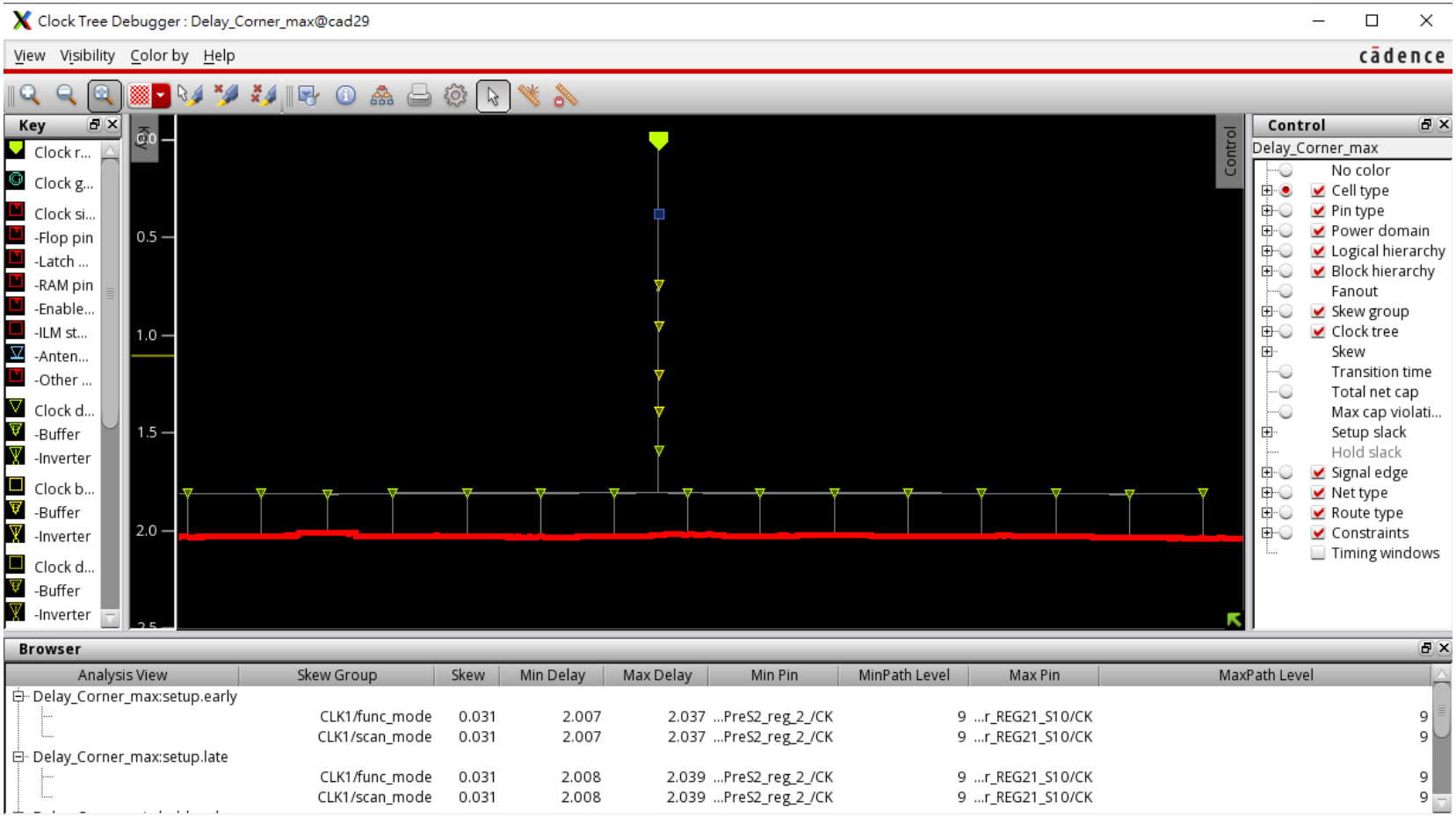


Create CCOpt Specification for CTS

- Create a clock tree specification by analyzing the timing graph structure of all active setup and hold analysis views
  - `Innovus #> create_ccopt_clock_tree_spec`
- Or written to a file for inspection and then loaded
  - `Innovus #> create_ccopt_clock_tree_spec -file ccopt.spec`
  - `Innovus #> source ccopt.spec`
- A clock tree specification contains clock_tree, skew_group and property settings

CCOpt Clock Tree Debugger

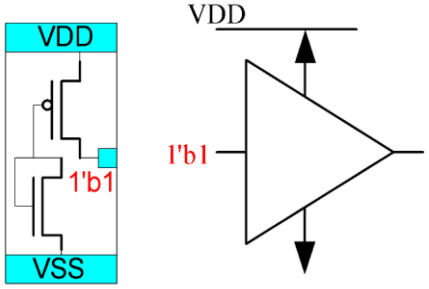
- ***Clock -> CCOpt Clock Tree Debugger***



Add Tiehi/Tielo Cell

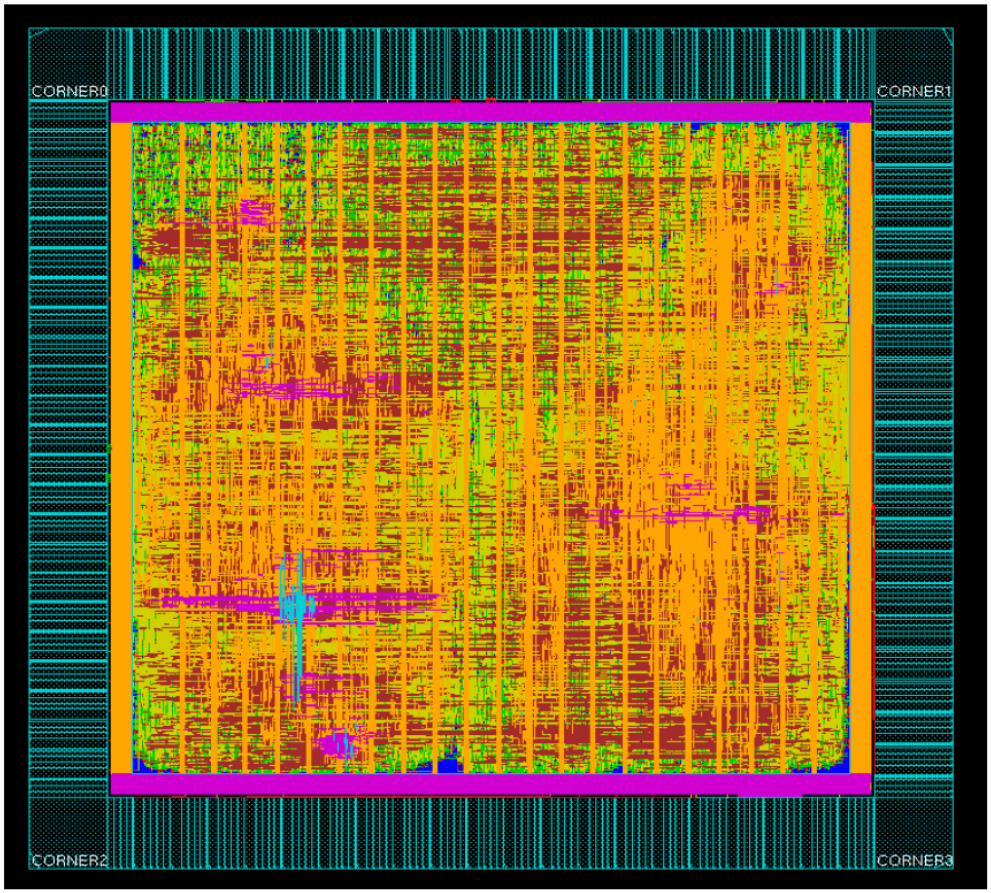
- Tiehi/Tielo cell connect tiehi/tielo net to supply voltage or ground with resister
- Tiehi/Tielo cell is added for ESD protection.
- ***Place -> Tie Hi/Lo Cell -> Add***



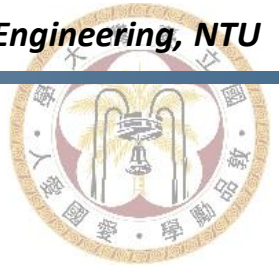


Routing

- **A process to add wires essential for properly connecting the placed components while following all design rules**
  - Std cells
  - Macro pins
  - I/O pads



- **Goals**
  - Optimized wire length
  - Min # of vias
  - Meet timing constraints
  - No LVS error
  - No DRC violations



After Routing

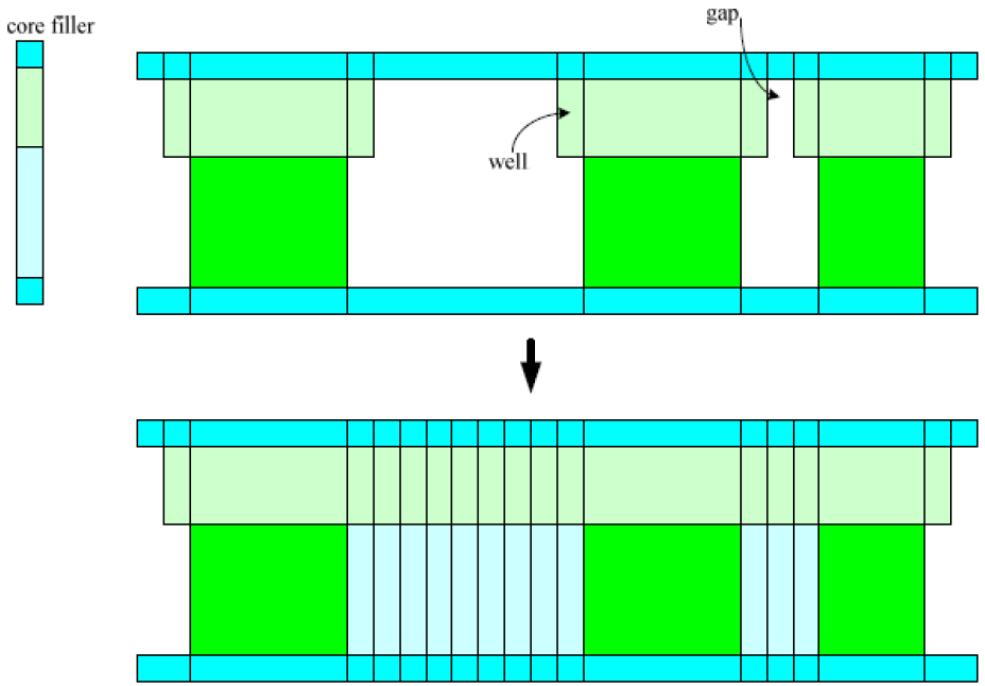
- **Still may have DRC violations since it uses tricks to prevent checking rules and faster the process**
  - ***"Search and Repair"*** after routing
  - Re-routing again
  - Use other tools to do DRC check

- **If DRC violations cannot be cleared in ~50 search & repair loops**
  - Change your floorplan or redesign your system

Add Core Filler

- **The core fillers are usually added to the place where std cells are not placed to make the density of the chip more uniform**
- ***Place -> Filler -> Add Filler***
  - Connect the NWELL/PWELL layer in core rows
  - Insert WELL contact

Add from wider filler to narrow filler (automatic)



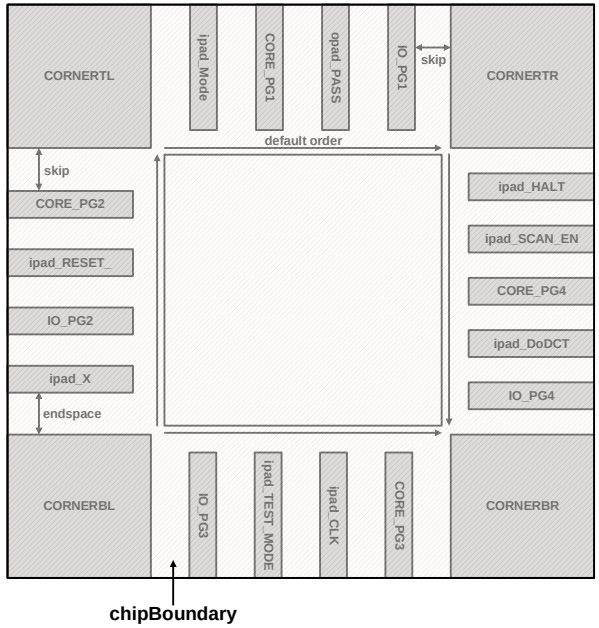
Chip Boundary

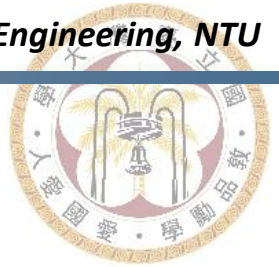
- One and only one chip boundary in a chip
- All tapeout layers must be inside chip boundary
- Generate chip boundary with custom shape

```
add_gui_shape -layer CUSTOM_CB -rect $cbox
```

- Add chip boundary layer to streamOut.map

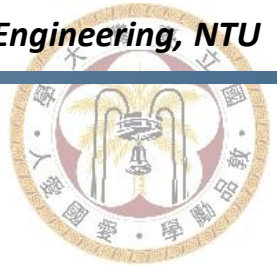
```
CUSTOM_CB CUSTOM 108 250
```





Output Data

- **Export GDS for DRC, LVS, LPE & tapeout**
  - ***Design -> Save -> GDS*** **LPE: layout parasitic extraction**
- **Export Netlist for LVS and simulation**
  - ***Design -> Save -> Netlist***
- **Export the netlist and the .sdf for post layout simulation**
  - `Innovus #> write_sdf`
- **Export DEF for reordered scan chain**
  - ***Design -> Save -> DEF***



Export .sdf

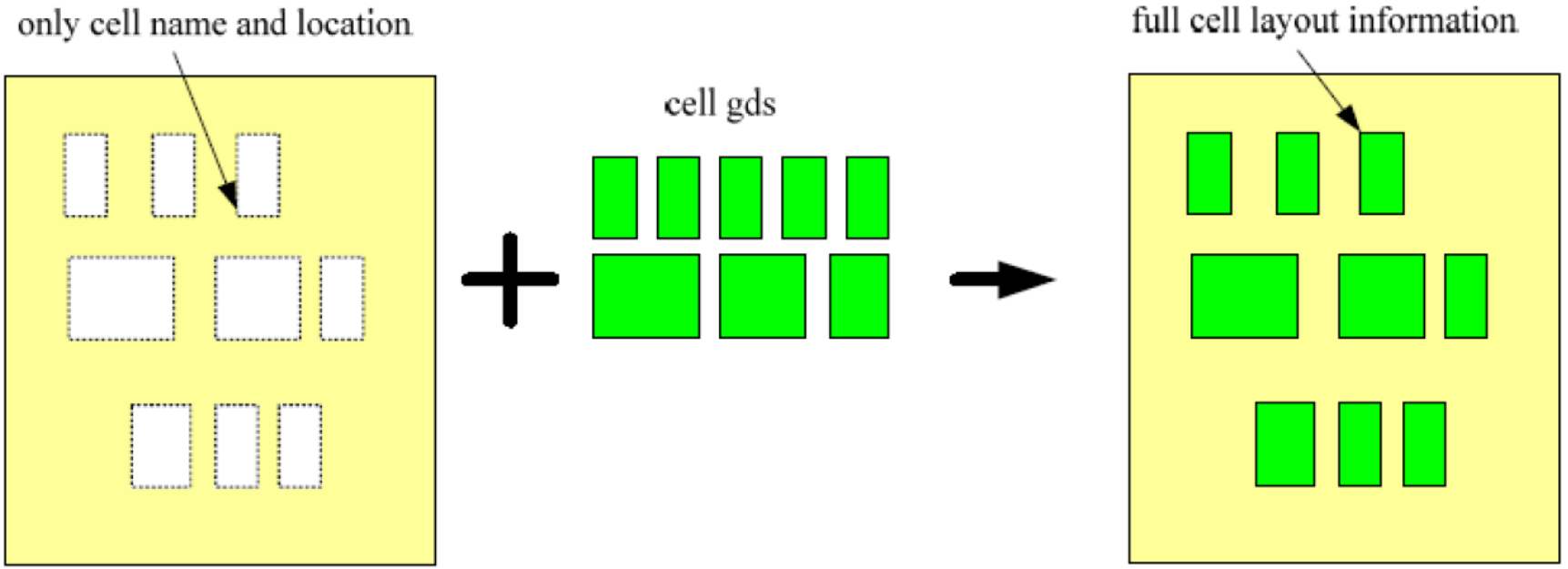
- **Use following commands to export .sdf**

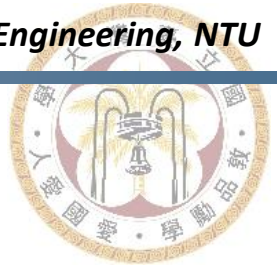
```
setAnalysisMode  -analysisType bcwc

write_sdf        -max_view AV_func_max \
                 -min_view AV_func_min \
                 -edges noedge \
                 -splitsetuphold \
                 -remashold \
                 -splitrecrem \
                 -min_period_edges none \
                  CHIP.sdf
```

What is GDS

- **Graphic data stream (GDS) is a binary file format for layout**
  - Represent layout data in a hierarchical format (.gds)
  - E.g. labels, shapes, layer information
  - Etching is based on the parameters provided in the file



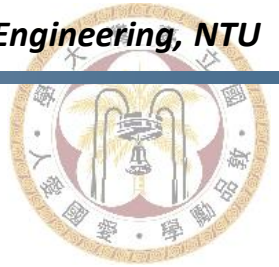


Export .gds

- **Use following commands to export .sdf:**

```
setStreamOutMode  -specifyViaName default \
                  -Sevianames false \
                  -virtualConnection false \
                  -uniquifyCellNamesPrefix false \
                  -snapToMGrid false \
                  -textSize 1\
                  -version 3

streamOut   CHIP.gds  -mapFile streamOut.map \
                      -merge {XXX.gds XXXX.gds} \
                      -stripes 1 -units 1000 -mode ALL
```

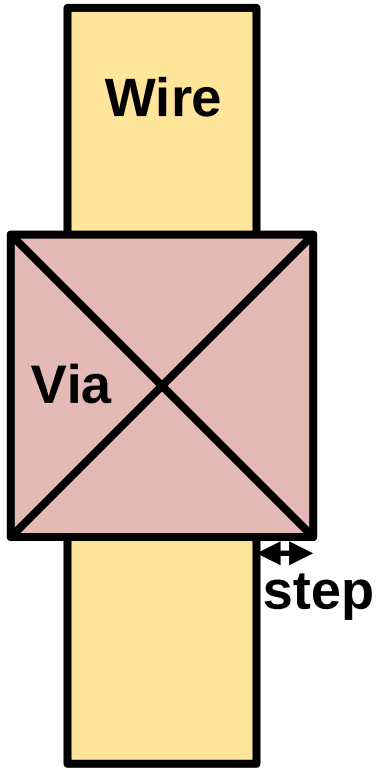


Outline

- Automatic Place & Route – Innovus
  - Power Planning
  - Placement
  - Clock Tree Synthesis
  - Routing
  - Output Data
- **Common DRC Errors**

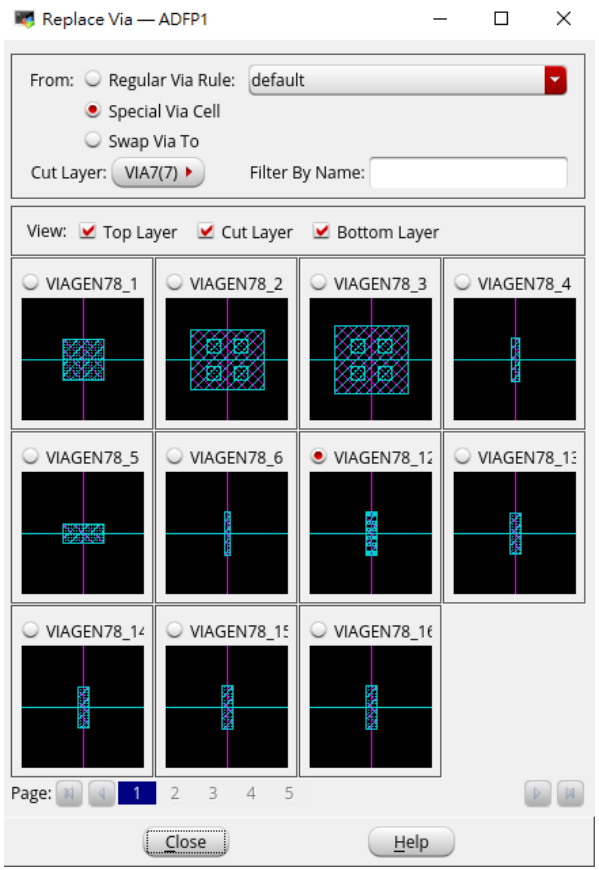
Fix Via DRC Errors

- Fix the following types of violations in special net via
- Short: `Innovus #> fixVia -short`
- Mincut: `Innovus #> fixVia -min_cut`
- Minstep: `Innovus #> fixVia -min_step`



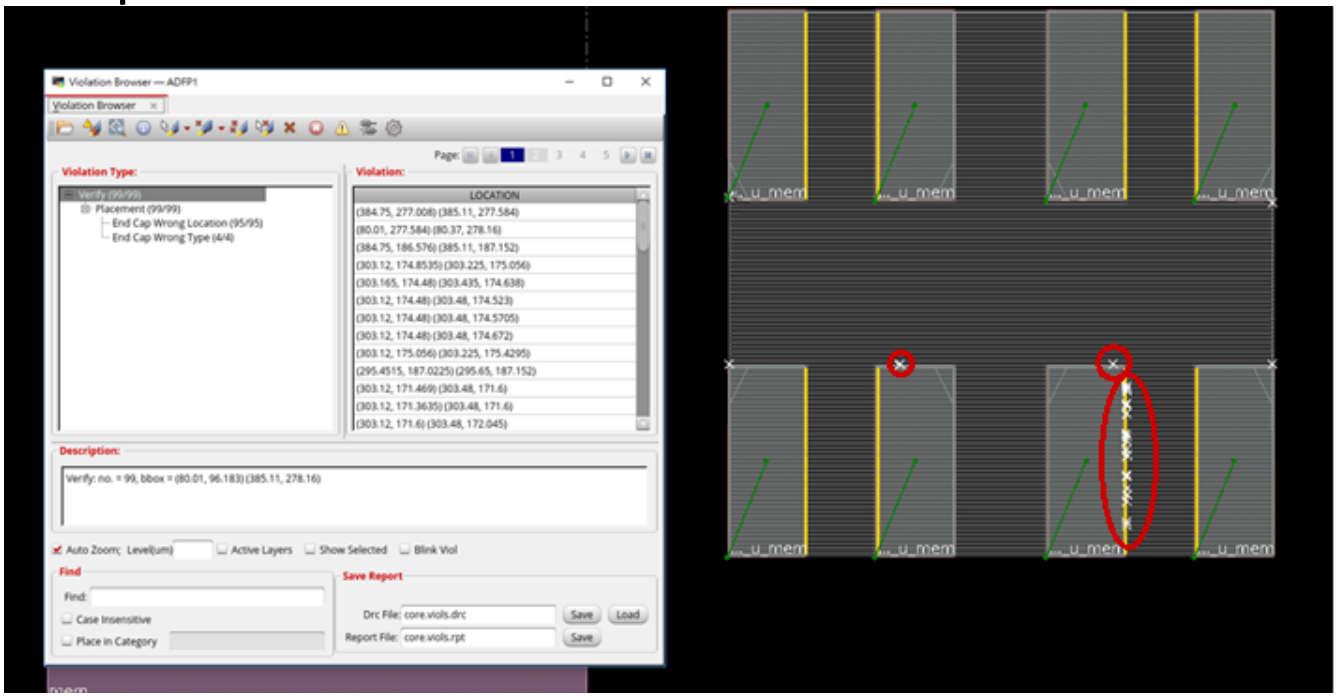
Via Replacement

- Manually replace a selected via with another via
  - Can be used when occurring via DRC error
- Press *F4*



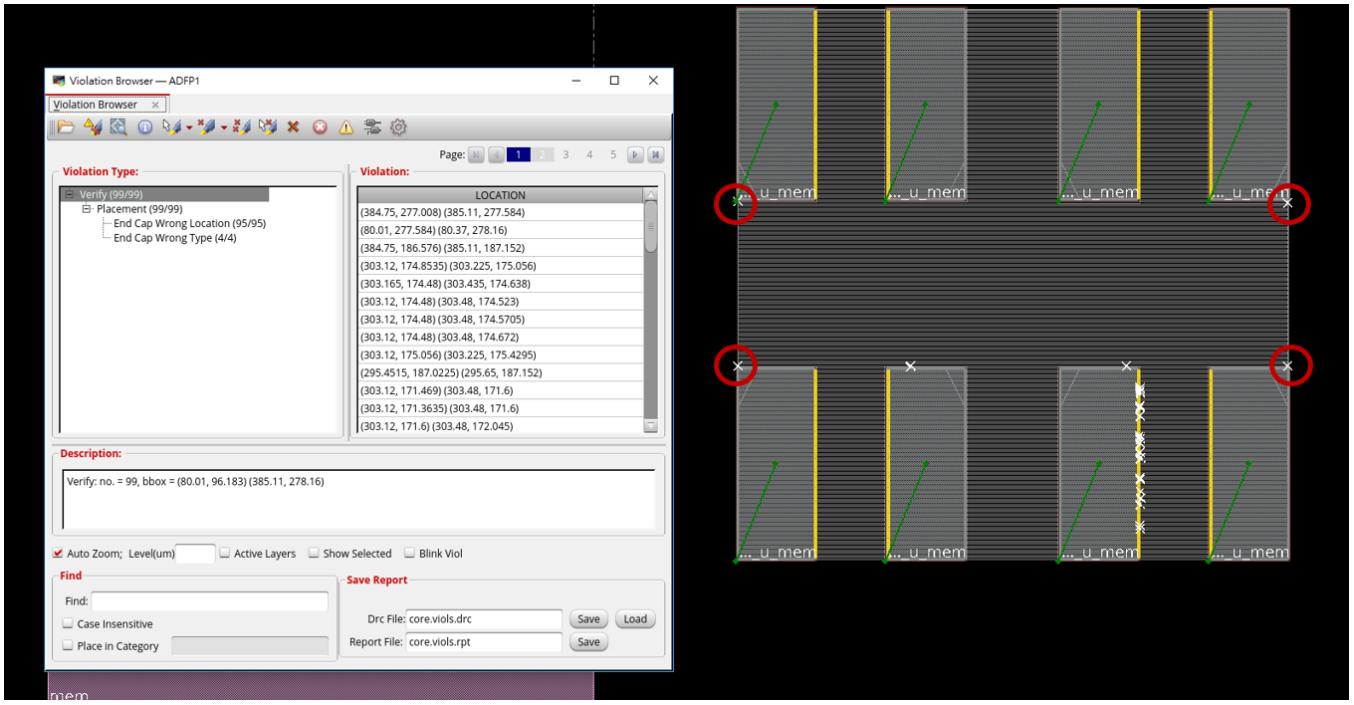
Endcap Wrong Location Error

- Endcap wrong location error: Endcap instance has overlap with other instance
  - You may use macros auto placement and forget to clear the standard cells from floorplan so that endcap cells have overlaps with standard cells



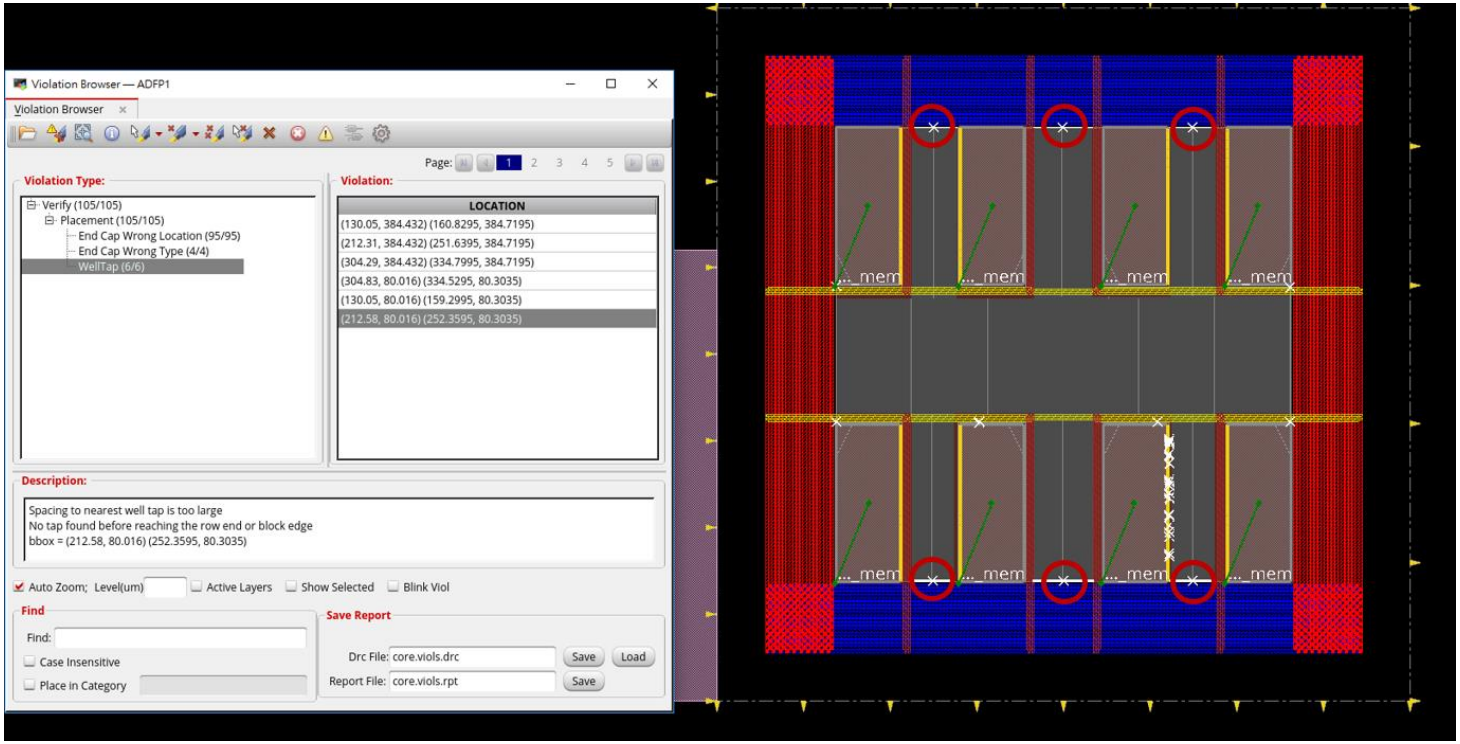
Endcap Wrong Type Error

- Endcap wrong type errors occur in the intersections of macros and boundaries
  - Case 1: Forget to set endcap type
  - Case 2: Forget to add placement blockage around macros



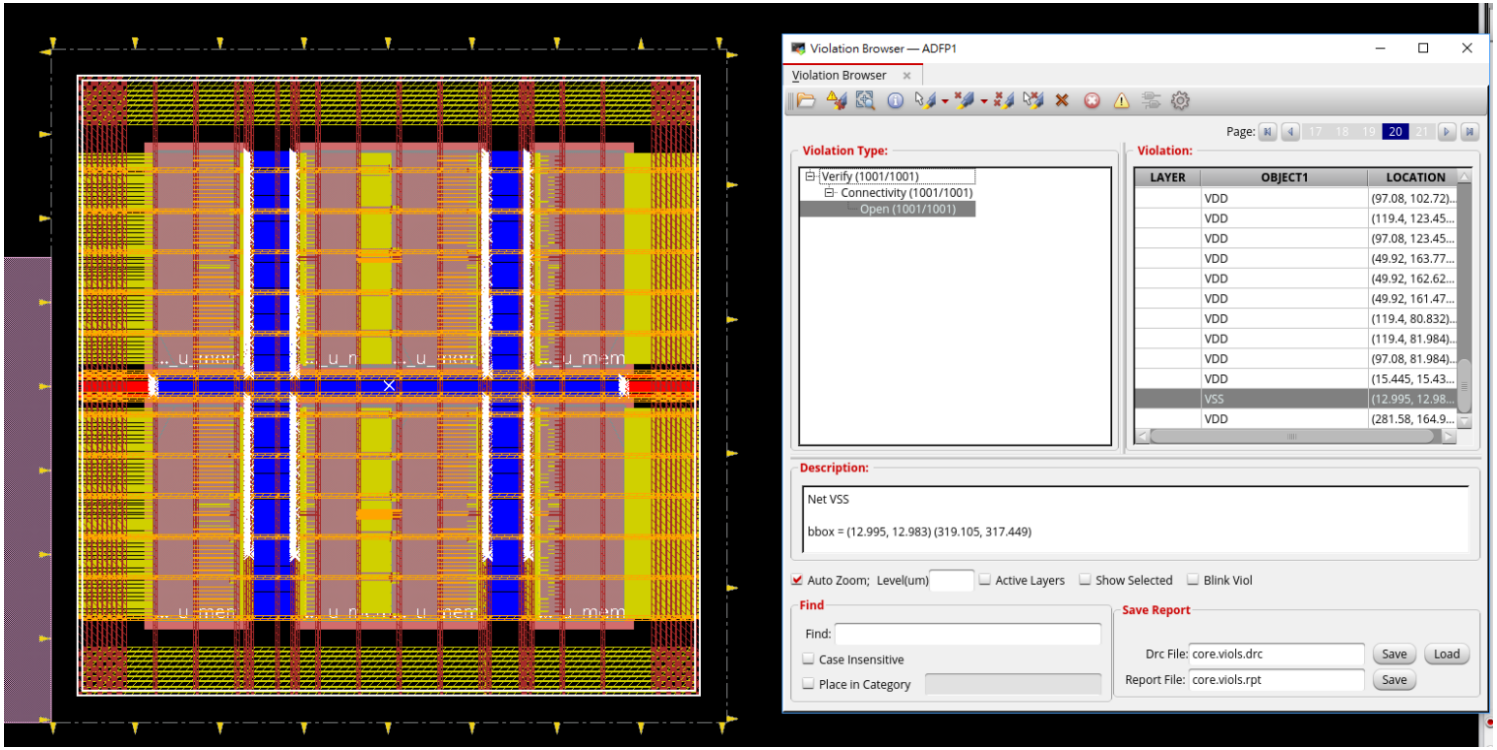
Well Tap Distance Error

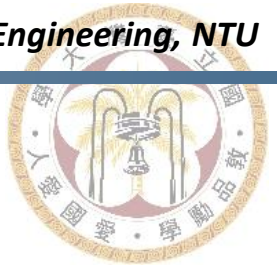
- Spacing to nearest well tap is too large
  - Case 1: Distance setting of well taps is too large
  - Case 2: If there is an endcap_tap cell nearby in the same row, it is a fake error



Connectivity Error

- Lots of open errors
  - Add core filler then verify connectivity, these errors should disappear





VDD/VSS Open on NW/PW Layer

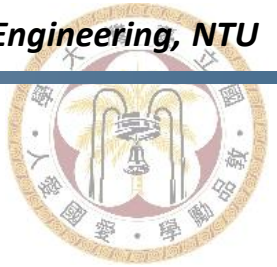
- Would find VDD open on NW layer and VSS open on PW layer when verify_connectivity
  - Modify standard cell lef file
  - Or directly ignore these errors

```
MACRO *TAPCELL*
    PIN VSS/VDD
        PORT
            LAYER M1;
                RECT …
        #END
        #PORT
            LAYER PW/NW;
                RECT …
        END
    END
```

VDDPST/ VSS/VDD Open on IO Ring

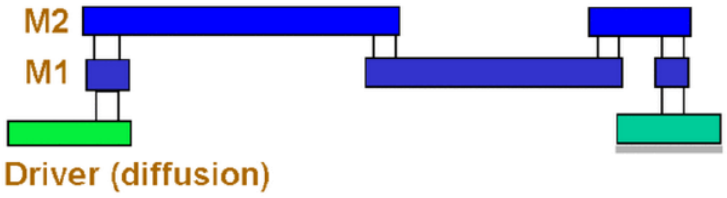
- Would find VDDPST/VSS/VDD open on IO ring
  - Modify I/O lef file
  - Or directly ignore these errors

```
MACRO P*
    PIN VDDPST
        PORT
            LAYER M5;
                RECT …
        #END
        #PORT
            LAYER M5;
                RECT …
        #END
        #PORT
            LAYER M6;
                RECT …
        #END
        #PORT
…
```



Antenna Effect

- In a chip manufacturing process, metal is initially deposited so it covers the entire chip
- The unneeded portions of the metal are removed by etching
- The exposed metal collect charge and form voltage potential
- If the voltage potential across the gate oxide becomes large enough, the current can damage the gate oxide
- Problem repair
  - Add jumper (change metal layer)
  - Add antenna cell (diode)
  - Add buffer



- No antenna cell in ADFP, just ignore antenna violations